



26

Which relation(s) will have no foreign key?
(0/2 Puan)

A , E

Doğru yanltlar: A, C, D



27

Using only the letter identifier for each entity, provide appropriate names for the relations mapped from the ER model.
(0/6 Puan)

A , B , C , E

Doğru yanltlar: A, B, C, CX (Representing the many-to-many relationship between C and X), XE (Representing combining X and E), D



28

If the cardinality for each relationship is changed to many-to-many with mandatory participation for all entities; how many relations would be derived from this version of the ER model?
(0/3 Puan)



1

Which task that is part of the LOGICAL design process is considered optional? Explain why this task is optional.

(0/10 Puan)

It is part of Merge. We don't need these when we don't have multiple logical design processes.

Doğru yanıtlar: Review with user because the user might not be available to meet

2

Explain the difference between SQL, VLSI, and SRAM

(-/10 Puan)

We haven't seen this thread

3

Define integrity constraints
(1/1 Puan)

☐ Conceptual Design

☒ Logical Design ✓

☐ Physical Design

4

Normalization
(1/1 Puan)

☐ Conceptual Design

☒ Logical Design ✓

☐ Physical Design

5

Determine key (CK, PK) attributes
(1/1 Puan)

☒ Conceptual Design ✓

☐ Logical Design

☐ Physical Design

6

Identify relationship types 
(1/1 Puan)

☒ Conceptual Design 

☐ Logical Design

☐ Physical Design

7

Design security mechanisms
(1/1 Puan)

☐ Conceptual Design

☐ Logical Design

☒ Physical Design 

8

The output of this phase is an E-R diagram
(1/1 Puan)

☒ Conceptual Design 

☐ Logical Design

☐ Physical Design



9

Merge local models into global model 
(0/1 Puan)

☐ Conceptual Design ✓

☒ Logical Design

☐ Physical Design



10

Review with the users
(0/1 Puan)

☒ Conceptual Design ✓

☒ Logical Design ✓

☐ Physical Design ✓

11

Choose indexes
(1/1 Puan)

☐ Conceptual Design

☐ Logical Design

☒ Physical Design ✓

12

Design user views
(1/1 Puan)

- ☐ Conceptual Design
- ☐ Logical Design
- ☒ Physical Design ✓

Normalization

Your team is tasked with designing the Student scheduling application for Courses in a university. Below are some example screens from the interface for a student. You will be asked questions based on how you would normalize a database for this system. NOTE: for this section, you are only designing the portion of the system that students will use.

13

Screen 1: Look Up Classes

Fall 2021

Department: Computer Engineering

1000 Personal Computer Applications

[View Sections](#)

1200 Intro to Computers

[View Sections](#)

1210 Fundamentals of Computing I

[View Sections](#)

Screen 2: Sections

Add	Dept	Course	Section	Credit	Title	Days	Time	Capacity	Actual	Rem	Instructor
☐	COMP	1210	1	3	Fundamentals of Computing I	MW	12:00-13:15	25	25	0	John Doe
☐	COMP	1210	2	3	Fundamentals of Computing I	MW	13:30-14:45	25	25	0	John Doe
☐	COMP	1210	3	3	Fundamentals of Computing I	MW	15:00-16:15	25	21	4	John Doe
☒	COMP	1210	4	3	Fundamentals of Computing I	MW	16:30-17:45	25	13	12	John Doe

Yanıtınızı girin

Screen 3: Grade Summary for Spring 2021 for Student Nusret Gökçe

Dept	Course	Section	Title	Grade	Credit	Earned	GPA points
COMP	3000	3	Object Oriented Programming	AA	3	3	12
COMP	3270	2	Intro to Algorithms	DD	3	3	3
COMP	3350	1	Computer Organization and Assembly	BB	3	3	9
COMP	3500	1	Intro to Operating Systems	FF	4	0	0
GPA Summary							
		Attempted	Earned	GPA Points	GPA		
Current Term		13	9	24	1.85		
Overall		80	71	184	2.30		

Yanıtınızı girin

16

Explain what tables you would create and for each, what would be the PK and FKs.
(-/15 Puan)

Dept(dept,deptName)
PrimaryKey - dept
Course(courseId,title,credit,dept)
PrimaryKey - courseId
ForeignKey - dept
Section(courseId,sectionId,days,time,capacity,actual,instructor)
PrimaryKey - courseId,sectionId
ForeignKey - courseId
derived(Rem can be calculated by capacity,actual)
Student(studentId,courseId,sectionId,grade)
PrimaryKey - studentId
ForeignKey - courseId,sectionId
derived (GPA,earned can be calculated by Grade)



17

Identify one example of derived data from the screens shown. Explain what the source data is and why the item you listed is considered derived.
(0/5 Puan)

Rem is derived data.We can find it from the difference between the capacity and the actual

Doğru yanıtlar: Examples could include Title, Credit, Earned, or GPA

18

We discussed two ways to represent derived data. For the item you wrote above, which of the two ways would you choose to represent the data and explain why.
(-/10 Puan)

1. As a data field, GPA must be updated anytime students pass the course
2. As a calculated field in a query.GPA must be calculated every time.

19

Write (in English) a possible request as user could ask for that would display data from multiple tables. Below that, type the SQL query you would write to perform the request.
(-/5 Puan)

```
Select * From Dept d
Inner join course c ON c.dept=d.dept
Inner join section s ON s.courseId=c.courseId
Inner join student st ON st.courseId= s.courseId and st.sectionId=s.sectionId
```

20

Write (in English) a possible request as user could ask for that would display data involving an aggregate function. Below that, type the SQL query you would write to perform the request.
(-/5 Puan)

```
I calculate student attempted credits
Select st.studentId,SUM(c.credit) From student st
Inner join section s ON st.courseId=s.courseId and st.sectionId=s.sectionId
Inner join course c ON c.courseId=s.courseId
Group By(st.studentId)
```

21

Write (in English) a possible request as user could ask for that would add data to a table.
Below that, type the SQL query you would write to perform the request.
(-/5 Puan)

Adding course

Insert into course (CourseId,Title,Credit,Dept) Values ("102","Programming",6,"CSE")

22

Write (in English) a possible request as user could ask for that would modify data in a table.
Below that, type the SQL query you would write to perform the request.
(-/5 Puan)

update student grade

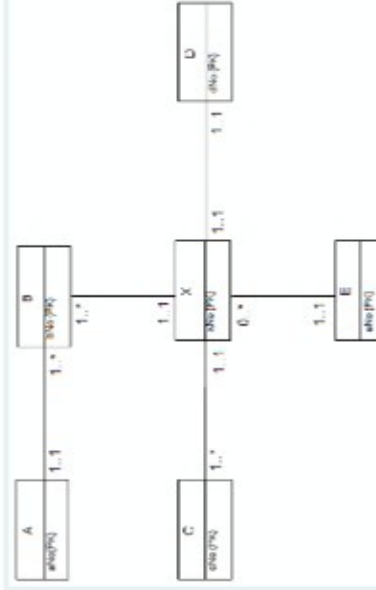
UPDATE student
SET grade= 'CC'
WHERE studentId= 1;

23

Write (in English) a possible request as user could ask for that would remove data from a table.
Below that, type the SQL query you would write to perform the request.
(-/5 Puan)

delete student datas from student table

DELETE FROM student
WHERE studentId= 1



Mapping

Answer the following questions with reference to how the ER model in the figure maps to relational tables.

24

How many relations will represent the ER model?
(0/2 Puan)

6

Doğru yanitlar: 6 relations

25

How many foreign keys will be in the relation representing X?
(0/2 Puan)

There is 1 foreign key E. (D is optional because it has one to one relationship)

Doğru yanitlar: 2 - The primary keys of entities B and D